

Edwin Kayang

Tempe, AZ epkayang@gmail.com +1 602 579 0159 Google Scholar www.github.com/pelpuo
www.pelpuo.github.io/ekayang-portfolio/

Education

Arizona State University <i>PhD Student in Computer Science</i>	<i>August 2023 – Present</i>
University of Ghana <i>BS in Computer Engineering</i> CGPA: 3.75/4.0	<i>Aug 2018 – Mar 2023</i>

Technologies

Languages: C, C++, Java, RISC-V Assembly, Python, JavaScript, Rust, Bash

Technologies: LLVM, Intel Pin, Tensorflow, PyTorch, Git, Cmake, GDB, Qemu, React/ReactNative, NodeJS, FastAPI, Flask, MLIR

Research Interests

Compilers, Program Analysis and Instrumentation, Profile Guided Optimization, System Security, Machine Learning, Fuzzing

Professional Experience

Research Assistant <i>Secure, Trusted and Assured Microelectronics (STAM) Center, Arizona State University</i>	<i>Tempe, AZ</i> <i>Aug 2023 – Present</i>
- Developing program analysis and compilation technologies aimed at ensuring memory safety in legacy applications, while maintaining performance. - Investigating algorithms to facilitate profile-guided optimization and compiler autotuning.	
Innovations Graduate Executive <i>4th-IR</i>	<i>Accra, Ghana</i> <i>June 2022 – Aug 2023</i>
- Developed and integrated application programming interfaces (APIs) for machine learning models into a model benchmarking platform. - Maintained local GPU servers for training, testing, and deploying machine learning models. - Conducted a comprehensive survey of machine learning models, providing rankings and recommendations for 20 distinct tasks.	
SuaCode Fellow <i>Nsesa Foundation</i>	<i>Remote</i> <i>June 2021 – Sept 2021</i>
- Contributed to a research paper detailing the design and implementation of an automated grading tool - Conducted testing and analysis of an automated grading tool	

Publications

Graphshot: A Large-Scale Program Representation Dataset for Machine Learning in Compilers <i>Edwin Kayang, Eric Jahns, Michel A. Kinsy</i> IEEE/ACM International Conference on Automated Software Engineering (Tools and Datasets Track)	<i>(Under Review)</i>
DALC-CT: Dynamic Analysis of Low-Level Code Traces for Constant-Time Verification Nges Brian Njungle, <i>Edwin Kayang</i> , Mishel Jyothis Paul, Michel A. Kinsy	<i>Oct 2025</i>

Panoptes: A Framework for Profile Clustering and Context-Aware Binary Optimization <i>(Just Accepted)</i> Edwin Kayang , Eric Jahns, Mishel Jyothis Paul, Michel A. Kinsy ACM/IEEE International Conference on Software Engineering	Dec 2025
Prismo: A Decision Support System for Privacy-Preserving ML Framework Selection Nges Brian Njungle, Eric Jahns, Luigi Mastromauro, Edwin Kayang , Milan Stojkov, Michel A. Kinsy arXiv preprint arXiv:2510.09985	Oct 2025
AQUILA: A Flexible Architecture Guideline for Building Custom Distributed Systems Testbeds <i>(Just Accepted)</i> Luigi Mastromauro, Edwin Kayang , Mishel Jyothis Paul, Eric Jahns, Muslum Ozgur Ozmen, Michel A. Kinsy IEEE/IFIP International Conference on Embedded and Ubiquitous Computing	Sept 2025
R-Visor: An Extensible Dynamic Binary Instrumentation and Analysis Framework for Open Instruction Set Architectures Edwin Kayang , Mishel Jyothis Paul, Eric Jahns, Muslum Ozgur Ozmen, Milan Stojkov, Kevin Rudd, Michel A. Kinsy ACM SIGPLAN/SIGBED Conference on Languages Compilers Tools and Theory for Embedded Systems doi.org/10.1145/3735452.3735522	June 2025
Autograd: Automated grading software for mobile game assignments in Suacode courses Prince Steven Annor, Edwin Kayang , Samuel Boateng, George Boateng Computer Science Education Research Conference doi.org/10.1145/3507923.3507954	Nov 2021

Invited Talks

RAIL: RISC-V Analysis and Instrumentation Library Latch-Up 2024, The FOSSi Foundation	Apr 2024
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Professional Affiliations

Tribe Lead, African PhD Students Network at ASU ◦ Organizing social events to connect PhD students and alumni of Arizona State University	Aug 2025 - Present
Shadow PC Reviewer, International Conference on Software Engineering (ICSE) ◦ Gave comprehensive reviews for 7 papers submitted to ICSE 2026	Aug 2025 - Oct 2025
Machine Learning, DevOps and 3D Modeling Instructor, Trestle Academy, Ghana ◦ Taught 3D modeling with Blender, Devops technologies such as Github actions and Machine learning with tensorflow to roughly 30 students.	June 2023 - Aug 2023
IT/Publications Head, Enactus, University of Ghana ◦ Led a team of web and mobile app developers to create solutions for the Sustainable Development Goals	Oct 2021 - June 2023
Lead, Google Developer Student Clubs, University of Ghana ◦ Organized tech events focused on teaching skills and exposing upcoming software developers to industry practices.	Oct 2020 - Aug 2021